

Subject : Physics,
Chemistry,
Biology

NEET GRAND Test 2

Paper Set : 1

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Total Mark : 720

Physics - Section A (MCQ)

- (1) A disc of mass m and radius r rolls down an inclined plane of height h . When it reaches the bottom of the plane, its rotational kinetic energy is (g = acceleration due to gravity).
- (A) $\frac{mgh}{3}$ (B) $\frac{mgh}{6}$
(C) $\frac{mgh}{2}$ (D) $\frac{mgh}{4}$
- (2) Out of the following statements, which one is 'NOT' correct in the case of a thermodynamic process?
- (A) In an isothermal process, $\Delta T = 0$
(B) In an isobaric process, $\Delta P = 0$
(C) In an isochoric process, $W = 0$
(D) In an isothermal process, $Q = 0$
- (3) The work done in splitting a water drop of radius R into 64 droplets is (T = Surface tension of water). (in πTR^2)
- (A) 6 (B) 12
(C) 8 (D) 24
- (4) *Assertion*: On a rainy day, it is difficult to drive a car or bus at high speed.
Reason: The value of the coefficient of friction is lowered due to the wetting of the surface.
- (A) If both *Assertion* and *Reason* are correct and the *Reason* is a correct explanation of the *Assertion*.
(B) If both *Assertion* and *Reason* are correct but *Reason* is not a correct explanation of the *Assertion*.
(C) If the *Assertion* is correct but *Reason* is incorrect.
(D) If both the *Assertion* and *Reason* are incorrect.
- (5) For a satellite moving in an orbit around the earth, the ratio of kinetic energy to potential energy is
- (A) 2 (B) $-1/2$
(C) $1/\sqrt{2}$ (D) $\sqrt{2}$
- (6) The length of the simple pendulum is made 3 times the original length. If T is its original time period, then the new time period will be:
- (A) $3T$ (B) $\sqrt{3}T$
(C) $\frac{T}{\sqrt{3}}$ (D) $\frac{T}{3}$
- (7) Energy is not carried by which of the following waves?
- (A) Stationary
(B) Progressive
(C) Transverse
(D) Electromagnetic
- (8) "Heat cannot by itself flow from a body at lower temperature to a body at higher temperature" is a statement or consequence of
- (A) Second law of thermodynamics
(B) Conservation of momentum
(C) Conservation of mass
(D) First law of thermodynamics
- (9) Preeti reached the metro station and found that the escalator was not working. She walked up the stationary escalator in time t_1 . On other days, if she remains stationary on the moving escalator, then the escalator takes her up in time t_2 . The time taken by her to walk up on the moving escalator will be
- (A) $\frac{t_1 t_2}{t_2 - t_1}$ (B) $\frac{t_1 t_2}{t_1 + t_2}$
(C) $t_1 - t_2$ (D) $\frac{t_1 + t_2}{2}$
- (10) 300 g of water at 25°C is added to 100 g of ice at 0°C . The final temperature of the mixture is (in $^\circ\text{C}$)
- (A) 12.5 (B) 0
(C) 25 (D) 50
- (11) A beaker full of hot water is kept in a room. If it cools from 80°C to 75°C in t_1 minutes, from 75°C to 70°C in t_2 minutes, and from 70°C to 65°C in t_3 minutes, then:
- (A) $t_1 = t_2 = t_3$ (B) $t_1 < t_2 = t_3$
(C) $t_1 < t_2 < t_3$ (D) $t_1 > t_2 > t_3$
- (12) The length of a metal wire is L_1 when the

tension is T_1 and L_2 when the tension is T_2 . The unstretched length of the wire is:

- (A) $\frac{L_1+L_2}{2}$
 (B) $\sqrt{L_1 L_2}$
 (C) $\frac{T_2 L_1 - T_1 L_2}{T_2 - T_1}$
 (D) $\frac{T_2 L_1 + T_1 L_2}{T_2 + T_1}$

(13) If the linear momentum is increased by 5%, the kinetic energy will increase by.....%

- (A) 5 (B) 10
 (C) 10.25 (D) 25

(14) A monoatomic gas is heated at constant pressure. The percentage of total heat used for doing external work is (in %)

- (A) 30 (B) 40
 (C) 50 (D) 60

(15) Identify the pair whose dimensions are equal.

- (A) Torque and work
 (B) Stress and energy
 (C) Force and stress
 (D) Force and work

(16) If the degree of freedom of a gas is f , then the ratio of two specific heats C_P/C_V is given by

- (A) $1 + \frac{2}{f}$ (B) $1 - \frac{2}{f}$
 (C) $1 + \frac{1}{f}$ (D) $1 - \frac{1}{f}$

(17) The magnitude of the sum of the two vectors $\vec{A}$ and $\vec{B}$ is equal to the magnitude of the difference of the two vectors $\vec{A}$ and $\vec{B}$. The angle between $\vec{A}$ and $\vec{B}$ is: (in $^\circ$)

- (A) 30 (B) 45
 (C) 90 (D) 180

(18) A vehicle travels half the distance with speed v and the remaining distance with speed $2v$. Its average speed is:

- (A) $\frac{3v}{4}$ (B) $\frac{v}{3}$
 (C) $\frac{2v}{3}$ (D) $\frac{4v}{3}$

(19) If the focal length of the objective lens is increased, then the magnifying power of:

- (A) microscope will increase but that of telescope will decrease.
 (B) microscope and telescope both will increase.
 (C) microscope and telescope both will decrease.
 (D) microscope will decrease but that of telescope will increase.

(20) Silicon and copper are cooled from 300 K to 100 K. What happens to their specific

resistance (resistivity)?

- (A) Increase in both copper and silicon
 (B) Decrease in both copper and silicon
 (C) Decrease in copper and increase in silicon
 (D) Increase in copper but decrease in silicon

(21) For a purely inductive or a purely capacitive circuit, the power factor is

- (A) zero (B) 0.5
 (C) 1 (D) ∞

(22) The self-inductance (L) produced by a solenoid of length ' l ' having ' N ' number of turns and cross-sectional area ' A ' is given by the formula (where ϕ = magnetic flux, μ_0 = permeability of vacuum).

- (A) $L = N\phi$ (B) $L = \mu_0 N A l$
 (C) $L = \frac{\mu_0 N^2 A}{l}$ (D) $L = \frac{\mu_0 N A}{l}$

(23) Choose the correct relation between polarisation P and electric susceptibility χ_e of a dielectric material. (E = electric field)

- (A) $P = \frac{\chi_e}{E^2}$ (B) $P = \frac{\chi_e}{E}$
 (C) $P = \chi_e E$ (D) $P = \chi_e^2 E$

(24) The correct relation between total magnetic field (B), magnetic intensity (H), permeability of free space (μ_0) and susceptibility (χ) is

- (A) $\frac{B}{H} = \mu_0(1 - \chi)$ (B) $\frac{B}{H} = \mu_0(1 + \chi)^2$
 (C) $\frac{B}{H} = \mu_0(1 + \chi)$ (D) $\frac{B}{H} = \mu_0(1 - \chi)^2$

(25) The ratio of contributions made by the electric field and magnetic field components to the intensity of an electromagnetic wave is:

- (A) 1 : c^2 (B) c : 1
 (C) 1 : 1 (D) 1 : c

(26) Electromagnetic waves are transverse in nature is evident by

- (A) Polarization
 (B) Interference
 (C) Reflection
 (D) Diffraction

(27) If ϵ_0 and ϵ represent the permittivity of free space and absolute permittivity of a medium, then the relative permittivity of the medium is:

- (A) $\frac{\epsilon^2}{\epsilon_0}$ (B) $\frac{\epsilon_0}{\epsilon}$
 (C) $\epsilon \epsilon_0$ (D) $\frac{\epsilon}{\epsilon_0}$

(28) In an interference experiment, the spacing between successive maxima or minima is

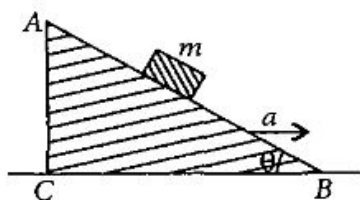
- (A) $\lambda d/D$ (B) $\lambda D/d$
 (C) dD/λ (D) $\lambda d/4D$

(29) A 42 mH inductor is connected to a 200 V, 50 Hz

- a.c. supply. The r.m.s. value of current in the circuit will be nearly (Take $\pi = \frac{22}{7}$) (in A)
- (A) 15.15 (B) 9.15
(C) 8.15 (D) 6.15
- (30) For a thin prism, if the angle of prism is 4° and the refractive index is 1.6, then the angle of minimum deviation will be (in $^\circ$)
- (A) 1.6 (B) 2.0
(C) 2.4 (D) 0.4
- (31) According to Maxwell's hypothesis, a changing electric field gives rise to
- (A) An e.m.f. (B) Electric current
(C) Magnetic field (D) Pressure radiant
- (32) Magnetic permeability is maximum for
- (A) Diamagnetic substance
(B) Paramagnetic substance
(C) Ferromagnetic substance
(D) All of these
- (33) An electric kettle has two heating coils. When one of the coils is connected to an a.c. source, the water in the kettle boils in 10 minutes. When the other coil is used, the water boils in 40 minutes. If both the coils are connected in parallel, the time taken by the same quantity of water to boil will be min.
- (A) 15 (B) 25
(C) 8 (D) 4
- (34) Two coherent sources of light interfere and produce a fringe pattern on a screen. For the central maximum, the phase difference between the two waves will be,
- (A) $\frac{\pi}{2}$ (B) 0
(C) π (D) $\frac{3\pi}{2}$
- (35) A charge Q is placed at the centre of a cube of side a . The total flux of electric field through the six surfaces of the cube is
- (A) $\frac{6Qa^2}{\epsilon_0}$ (B) $\frac{Qa^2}{6\epsilon_0}$
(C) Q/ϵ_0 (D) Qa^2/ϵ_0
- (36) Assertion: The resolving power of a telescope is more if the diameter of the objective lens is more.
Reason: Objective lens of large diameter collects more light.
- (A) If both Assertion and Reason are correct and the Reason is a correct explanation of the Assertion.
(B) If both Assertion and Reason are correct but Reason is not a correct explanation of the Assertion.
(C) If the Assertion is correct but Reason is incorrect.
(D) If both the Assertion and Reason are incorrect.
- (37) Radioactive nuclei that are injected into a patient collect at certain sites within their body, undergoing radioactive decay and emitting electromagnetic radiation. These radiations can then be recorded by a detector. This procedure provides an important diagnostic tool called:
- (A) Gamma camera
(B) CAT scan
(C) Radiotracer technique
(D) Gamma ray spectroscopy
- (38) The refractive index of water is 1.33. What will be the speed of light in water?
- (A) $3 \times 10^8 \text{ m/s}$ (B) $2.25 \times 10^8 \text{ m/s}$
(C) $4 \times 10^8 \text{ m/s}$ (D) $1.33 \times 10^8 \text{ m/s}$
- (39) A metal surface is illuminated by light of given intensity and frequency to cause photoemission. If the intensity of illumination is reduced to one fourth of its original value, then the maximum KE of the emitted photoelectrons would be
- (A) Twice the original value
(B) Four times the original value
(C) One fourth of the original value
(D) Unchanged
- (40) Given the value of Rydberg constant is 10^7 m^{-1} , the wave number of the last line of the Balmer series in the hydrogen spectrum will be:
- (A) $0.5 \times 10^7 \text{ m}^{-1}$ (B) $0.25 \times 10^7 \text{ m}^{-1}$
(C) $2.5 \times 10^7 \text{ m}^{-1}$ (D) $0.025 \times 10^4 \text{ m}^{-1}$
- (41) The current associated with an electron of charge e moving in a circular orbit of radius r with speed v around the positively charged nucleus is given by
- (A) $\frac{er}{2\pi v}$ (B) $\frac{e\pi}{2rv}$
(C) $\frac{ev}{\pi r}$ (D) $\frac{ev}{2\pi r}$
- (42) If the initial tension on a stretched string is doubled, then the ratio of the initial and final speeds of a transverse wave along the string is:
- (A) $\sqrt{2} : 1$ (B) $1 : \sqrt{2}$
(C) $1 : 2$ (D) $1 : 1$

- (43) If the magnitude of the sum of two vectors is equal to the magnitude of the difference of the two vectors, the angle between these vectors is^o.
 (A) 90 (B) 120
 (C) 45 (D) 60

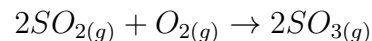
- (44) A block of mass m is placed on a smooth inclined wedge ABC of inclination θ as shown in the figure. The wedge is given an acceleration a towards the right. The relation between a and θ for the block to remain stationary on the wedge is



- (A) $a = g \csc \theta$ (B) $a = g \sin \theta$
 (C) $a = g \tan \theta$ (D) $a = g \cos \theta$
- (45) A disc and a sphere of same radius but different masses roll off on two inclined planes of the same altitude and length. Which one of the two objects gets to the bottom of the plane first?
 (A) Sphere
 (B) Both reach at the same time
 (C) Depends on their masses
 (D) Disc

Chemistry - Section A (MCQ)

- (46) What is the vapour density of O_2 gas?
 (A) 8 (B) 16
 (C) 32 (D) 22.4
- (47) Which among the following compounds exhibits +2 oxidation state of oxygen?
 (A) H_2O (B) SO_2
 (C) OF_2 (D) H_2O_2
- (48) Which of the following rules or principles states that "No two electrons in an atom can have the same set of four quantum numbers"?
 (A) Pauli's exclusion principle
 (B) Hund's rule
 (C) Aufbau rule
 (D) Heisenberg uncertainty principle
- (49) Calculate the mass in kg of 4.48 dm^3 of carbon dioxide (CO_2) at STP.
 (A) 2.2×10^{-3} (B) 4.4×10^{-3}
 (C) 6.6×10^{-3} (D) 8.8×10^{-3}
- (50) For the following reaction, the relation between ΔH and ΔU is:



- (A) $\Delta H = \Delta U - 2RT$ (B) $\Delta H = \Delta U - RT$
 (C) $\Delta H = \Delta U + 2RT$ (D) $\Delta H = \Delta U + RT$
- (51) What is the number of electrons around sulfur in H_2SO_4 molecule?
 (A) 4 (B) 6
 (C) 10 (D) 12
- (52) The reaction quotient Q for the reaction $N_{2(g)} + 3H_{2(g)} \rightleftharpoons 2NH_{3(g)}$ is given by $Q = \frac{[NH_3]^2}{[N_2][H_2]^3}$. The reaction will proceed from right to left when:
 (A) $Q = 0$ (B) $Q = K_c$
 (C) $Q < K_c$ (D) $Q > K_c$
- (53) Which of the following atoms should have the highest 1st electron affinity?
 (A) F (B) O
 (C) N (D) C
- (54) How many isomers are possible for an alkane having molecular formula C_5H_{12} ?
 (A) 5 (B) 3
 (C) 4 (D) 2
- (55) Identify the extensive property amongst the following.
 (A) Viscosity
 (B) Heat capacity
 (C) Density
 (D) Surface tension
- (56) Isomers have essentially identical
 (A) Structural formula
 (B) Chemical properties
 (C) Molecular formula
 (D) Physical properties
- (57) Which of the following formulae is used to find the $[OH^-]$ ion concentration of a weak monoacidic base?
 (A) $K_b \cdot c$ (B) $\sqrt{K_b \cdot c}$
 (C) $\sqrt{\frac{K_b}{c}}$ (D) $\sqrt{K_b}$
- (58) Which of the following functional groups has the lowest priority for deciding the principal functional group in a polyfunctional compound?
 (A) $-COOR$ (B) $-COCl$
 (C) $>C=O$ (D) $-CONH_2$
- (59) How many structural isomers are possible for a compound with molecular formula C_3H_7Cl ?
 (A) 2 (B) 5
 (C) 7 (D) 9
- (60) During Lassaigne's test, the elements present in an organic compound are converted from :

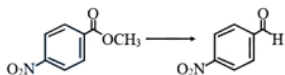
- (A) ionic form to ionic form
(B) covalent form to ionic form
(C) ionic form to covalent form
(D) covalent form to covalent form
- (61) In the acetylene molecule, the $C-H$ sigma bond is formed by the overlap of which orbitals?
(A) $s-p$ overlap (B) $sp-s$ overlap
(C) $p-p$ overlap (D) $sp-sp$ overlap
- (62) What is de Broglie's wavelength for a particle having mass $6.64 \times 10^{-27} \text{ kg}$ moving with velocity of $3 \times 10^3 \text{ ms}^{-1}$ (in nm)?
[$h = 6.63 \times 10^{-34} \text{ Js}$]
(A) 0.111 (B) 0.0333
(C) 1.22 (D) 2.42
- (63) Which amongst the following is the most stable carbocation?
(A) $CH_3-CH^+(CH_3)$ (B) $CH_3-C^+(CH_3)_2$
(C) CH_3^+ (D) $CH_3-CH_2^+$
- (64) One atomic mass unit (amu) is equal to:
(A) $1.66 \times 10^{-27} \text{ g}$ (B) $1.66 \times 10^{-24} \text{ g}$
(C) $1.66 \times 10^{-23} \text{ g}$ (D) $1.66 \times 10^{-25} \text{ kg}$
- (65) Which of the following is the most stable carbocation?
(A) CH_3^+ (B) $CH_3CH_2^+$
(C) $(CH_3)_2CH^+$ (D) $(CH_3)_3C^+$
- (66) Benzene reacts with CH_3COCl in the presence of $AlCl_3$ to give
(A) C_6H_5Cl (B) C_6H_5COCl
(C) $C_6H_5CH_3$ (D) $C_6H_5COCH_3$
- (67) Geometrical isomerism is shown by
(A) 2-butene (B) 2-butyne
(C) 2-butanol (D) Butanal
- (68) The reagent commonly used to determine the hardness of water titrimetrically is
(A) Oxalic acid
(B) Disodium salt of $EDTA$
(C) Sodium citrate
(D) Sodium thiosulphate
- (69) Which of the following enzymes is useful in the conversion of glucose to fructose?
(A) Invertase (B) Glucose isomerase
(C) Amylase (D) Proteases
- (70) Identify the name of the following reaction:
$$\text{Toluene} + \text{CrO}_2\text{Cl}_2 \xrightarrow{\text{CS}_2} \text{complex}$$

$$\xrightarrow{\text{H}_3\text{O}^+} \text{Benzaldehyde}$$
- (A) Stephen reaction
(B) Rosenmund reaction
(C) Etard reaction
(D) Wolf-Kishner reaction
- (71) Which one of the elements with the following outer orbital configurations may exhibit the largest number of oxidation states?
(A) $3d^5 4s^1$ (B) $3d^5 4s^2$
(C) $3d^2 4s^2$ (D) $3d^3 4s^2$
- (72) Which one of the following on reduction with lithium aluminium hydride yields a secondary amine?
(A) Methyl isocyanide (B) Acetamide
(C) Methyl cyanide (D) Nitroethane
- (73) The resistance of a conductivity cell filled with 0.1 M KCl solution is 100Ω and the conductivity is $1.70 \times 10^{-4} \text{ S cm}^{-1}$. What is the cell constant of the cell (in cm^{-1})?
(A) 1.70 (B) 0.017
(C) 0.058 (D) 0.012
- (74) Which among the following colours is obtained in Schiff test of aldehydes?
(A) Blue (B) Green
(C) Magenta (D) Black
- (75) Alkyl cyanides on reduction by sodium and ethanol give primary amines. This reaction is called as
(A) Wolff-Kishner reduction
(B) Hell-Volhard-Zelinsky reaction
(C) Mendius reduction
(D) Clemmensen reduction
- (76) A first order reaction takes 40 min for 20% decomposition. Calculate its rate constant.
(A) $5.6 \times 10^{-3} \text{ min}^{-1}$ (B) $4.5 \times 10^{-3} \text{ min}^{-1}$
(C) $6.5 \times 10^{-3} \text{ min}^{-1}$ (D) $7.2 \times 10^{-3} \text{ min}^{-1}$
- (77) The relative lowering of the vapour pressure is equal to the ratio between the number of
(A) Solute molecules and solvent molecules
(B) Solute molecules and the total molecules in the solution
(C) Solvent molecules and the total molecules in the solution
(D) Solvent molecules and the total number of ions of the solute
- (78) $EDTA$ (Ethylenediaminetetraacetate ion) is a hexadentate ligand. Identify the number of donor atoms in an $EDTA$ molecule that form coordinate bonds with a central metal atom or

ion in a complex.

- (A) 3 (B) 1
(C) 6 (D) 4

- (79) Which catalyst is used in the conversion of chlorobenzene to phenol by the Raschig process?
(A) Calcium phosphate (B) Calcium carbonate
(C) Calcium sulphate (D) Calcium chloride
- (80) During electrolysis of water, the volume of O_2 liberated is 2.24 dm^3 . The volume of hydrogen liberated, under same conditions, will be dm^3 .
(A) 2.24 (B) 1.12
(C) 4.48 (D) 0.56
- (81) The compound formed as a result of oxidation of ethyl benzene by $KMnO_4$ is
(A) benzyl alcohol (B) benzophenone
(C) acetophenone (D) benzoic acid
- (82) Identify the reagent used in the following conversion.



- (A) $CO, HCl/AlCl_3$ (B) $CrO_2Cl_2 \cdot H_3O^+$
(C) $AlH(i-Bu)_2, H_3O^+$ (D) $CrO_3, (CH_3CO)_2O$
- (83) The instantaneous rate for the reaction $2A + B \rightarrow C + 3D$ is given by:
(A) $\frac{1}{3} \frac{d[D]}{dt}$ (B) $-\frac{1}{2} \frac{d[A]}{dt}$
(C) $-\frac{d[A]}{dt}$ (D) $-\frac{d[B]}{dt}$
- (84) What is the shape and magnetic nature of the permanganate ion (MnO_4^-)?
(A) Pyramidal, diamagnetic
(B) Tetrahedral, diamagnetic
(C) Tetrahedral, paramagnetic
(D) Planar, paramagnetic
- (85) The conversion of ethyl bromide to ethyl iodide using sodium iodide and dry acetone is known as:
(A) Swarts reaction
(B) Finkelstein reaction
(C) Sandmeyer reaction
(D) Stephen reaction
- (86) Which of the following electronic configurations represents a transition element?
(A) $1s^2, 2s^2 2p^6, 3s^2 3p^6 3d^{10}, 4s^2 4p^6$
(B) $1s^2, 2s^2 2p^6, 3s^2 3p^6 3d^{10}, 4s^2 4p^1$
(C) $1s^2, 2s^2 2p^6, 3s^2 3p^6 3d^2, 4s^2$
(D) $1s^2, 2s^2 2p^6, 3s^2 3p^6, 4s^2$

- (87) What will be the molality of a solution having 18 g of glucose (mol. wt. = 180) dissolved in 500 g of water?

- (A) 1 (B) 0.5
(C) 0.2 (D) 2

- (88) Identify the product P of the following reaction:
Ethanoyl chloride $\xrightarrow{H_2O}$ P

- (A) Ethyl ethanoate (B) Ethanoic acid
(C) Ethanol (D) Ethanal

- (89) The presence or absence of a hydroxyl group on which carbon atom of the sugar differentiates RNA and DNA ?

- (A) 1^{st} (B) 2^{nd}
(C) 3^{rd} (D) 4^{th}

- (90) The magnetic moment (spin only) of $[NiCl_4]^{2-}$ is $B.M.$

- (A) 1.82 (B) 5.46
(C) 2.82 (D) 1.41

Biology - Section A (MCQ)

- (91) How many ATP molecules are formed in one glycolytic pathway?

- (A) 3 (B) 4
(C) 6 (D) 1

- (92) The characteristic feature of brown algae is the presence of

- (A) Phycocyanin (B) Phycoerythrin
(C) Fucoxanthin (D) Haematochrome

- (93) During the first meiotic division, the exchange of genetic material between the chromatids of homologous chromosomes occurs. What is this process called?

- (A) Transformation
(B) Chiasmata
(C) Crossing over
(D) Synapsis

- (94) Which of the following statements is correct regarding human respiration?

- (A) Cigarette smoking causes inflammation of the bronchi.
(B) Neural signals from the pneumotaxic center in the pons region of the brain increase the duration of inspiration.
(C) Workers in grinding and stone-breaking industries suffer from lung fibrosis.

- (D) (A) and (C)

- (95) Which one of the following is a matching pair?

- (A) Lubb - Sharp closure of *AV* valves at the beginning of ventricular systole.
 (B) Dup - Sudden opening of semilunar valves at the beginning of ventricular diastole.
 (C) Pulsation of the radial artery valves in the blood vessels.
 (D) Purkinje fibres - Initiation of the heart beat.
- (96) Each human kidney has nearly
 (A) 10,000 nephrons (B) 50,000 nephrons
 (C) 1,00,000 nephrons (D) 1 million nephrons
- (97) *ATP* synthesis occurs on the
 (A) Outer membrane of mitochondrion
 (B) Inner membrane of mitochondrion
 (C) Matrix
 (D) None of the above
- (98) *Caenorhabditiselegans* is a
 (A) virus (B) cyanobacterium
 (C) fungus (D) nematode
- (99) Assertion : Hemodialysis can save and prolong the life of uremic patients.
 Reason : Waste products like urea can be removed from the blood by the process of hemodialysis.
 (A) If both Assertion and Reason are correct and the Reason is a correct explanation of the Assertion.
 (B) If both Assertion and Reason are correct but Reason is not a correct explanation of the Assertion.
 (C) If the Assertion is correct but Reason is incorrect.
 (D) If both the Assertion and Reason are incorrect.
- (100) Vascular cambium and cork cambium are examples of
 (A) Lateral meristem
 (B) Apical meristem
 (C) Elements of xylem and phloem
 (D) Intercalary meristem
- (101) Which of the following plants are vascular but seedless?
 (A) Algae (B) Bryophytes
 (C) Pteridophytes (D) Gymnosperms
- (102) Synaptonemal complex is formed during
 (A) Meiosis
 (B) Amitosis
 (C) Mitosis
 (D) Cytokinesis
- (103) *ECG* helps in the diagnosis of the following abnormalities in conducting pathways *EXCEPT*
 (A) enlargement of heart chambers
 (B) damage to cardiac muscles
 (C) reduction in blood supply to cardiac muscles
 (D) leucopenia
- (104) Bulliform cells in grass leaves show
 (A) Growth movements
 (B) Tropic movements
 (C) Nastic movements
 (D) Turgor movements
- (105) All living members of the class Cyclostomata are:
 (A) Free living (B) Endoparasite
 (C) Symbiotic (D) Ectoparasite
- (106) Middle lamella is made up of
 (A) Cellulose
 (B) Suberin
 (C) Calcium and magnesium pectate
 (D) Lignin
- (107) In which one of the following do male and female gametophytes not have a free-living independent existence?
 (A) Polytrichum (B) Cedrus
 (C) Pteris (D) Funaria
- (108) Match column *I* with column *II* for housefly classification and select the correct option using the codes given below.
- | Column-I | Column-II |
|------------|-----------------|
| (A) Family | (i) Diptera |
| (B) Order | (ii) Arthropoda |
| (C) Class | (iii) Muscidae |
| (D) Phylum | (iv) Insecta |
- (A) *A – (iii), B – (i), C – (iv), D – (ii)*
 (B) *A – (iii), B – (ii), C – (iv), D – (i)*
 (C) *A – (iv), B – (iii), C – (ii), D – (i)*
 (D) *A – (iv), B – (ii), C – (i), D – (iii)*
- (109) Which of the following does *NOT* contribute to the formation of the thoracic cage?

- (A) Diaphragm
- (B) Sternum
- (C) Pleura
- (D) Ribs

(110) The number of floating ribs in humans is
 . (in *pairs*)

- (A) 2
- (B) 3
- (C) 5
- (D) 6

(111) What is the response of various organisms to the duration of light and dark periods called?

- (A) Phototropism
- (B) Phototaxis
- (C) Vernalization
- (D) Photoperiodism

(112) A marine cartilaginous fish that can produce electric current is

- (A) *Pristis*
- (B) *Torpedo*
- (C) *Trygon*
- (D) *Scoliodon*

(113) Which element is a constituent of all amino acids?

- (A) Copper
- (B) Phosphorus
- (C) Nitrogen
- (D) Magnesium

(114) Which of the following statements is incorrect?

- (A) Viroids lack a protein coat.
- (B) Viruses are obligate parasites.
- (C) Infective constituent in viruses is the protein coat.
- (D) Prions consist of abnormally folded proteins.

(115) The enzyme required for carboxylation in the Calvin cycle is:

- (A) (1) *PEP* carboxylase
- (B) (2) *RuBP* carboxylase-oxygenase
- (C) (3) Carboxypeptidase
- (D) (4) Hexokinase

(116) Given below are two statements:

Statement *I* : Amino acids have a property of ionizable nature of $-NH_2$ and $-COOH$ groups, hence have different structures at different *pH*.

Statement *II* : Amino acids can exist as Zwitterionic form at acidic and basic *pH*.

In the light of the above statements, choose the most appropriate answer from the options given below:

(A) Both Statement *I* and Statement *II* are correct

(B) Both Statement *I* and Statement *II* are incorrect

(C) Statement *I* is correct but Statement *II* is incorrect

(D) Statement *I* is incorrect but Statement *II* is correct

(117) A pair of hormones produced by the kidneys is:

- (A) Erythropoietin and Relaxin
- (B) Erythropoietin and Calcitriol
- (C) Calcitonin and Relaxin
- (D) Calcitonin and Calcitriol

(118) Singer and Nicholson's model of plasma membrane differs from Robertson's model in the

- (A) Number of lipid layers
- (B) Arrangement of lipid layers
- (C) Arrangement of proteins
- (D) Absence of protein in Singer and Nicholson's model

(119) One very special feature in the earthworm (*Pheretima*) is that

- (A) fertilization of eggs occurs inside the body
- (B) the typhlosole greatly increases the effective absorption area of the digested food in the intestine
- (C) the *S*-shaped setae embedded in the integument are the defensive weapons used against the enemies
- (D) it has a long dorsal tubular heart.

(120) The posterior pituitary gland is not a 'true' endocrine gland because

- (A) it is provided with a duct
- (B) it only stores and releases hormones
- (C) it is under the regulation of hypothalamus
- (D) it secretes enzymes.

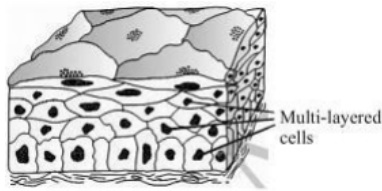
(121) The toxic effect of carbon monoxide is due to its greater affinity for haemoglobin as compared to oxygen by (approx.) (in *times*)

- (A) 200
- (B) 2
- (C) 1000
- (D) 20

(122) $K_{2+2}C_4A_{2+4}G_{(2)}$ is the floral formula of:

- (A) *Allium cepa*
- (B) *Solanum nigrum*
- (C) *Helianthus annuus*
- (D) *Brassica nigra*

(123) Identify the diagram and its related function.



- (A) Simple cuboidal epithelium - Diffusion
- (B) Simple squamous epithelium - Secretion and absorption
- (C) Compound epithelium - Protection
- (D) Compound epithelium - Diffusion

(124) Viruses have

- (A) *DNA* enclosed in a protein coat
- (B) prokaryotic nucleus
- (C) single chromosome
- (D) both *DNA* and *RNA*

(125) *ICBN* stands for

- (A) International Code of Botanical Nomenclature
- (B) International Congress of Biological Names
- (C) Indian Code of Botanical Nomenclature
- (D) Indian Congress of Biological Names

(126) Which one of the following types of reaction occurs during the conversion of isocitrate $\rightarrow$ oxalosuccinate?

- (A) Oxidation
- (B) Decarboxylation
- (C) Hydration
- (D) Dephosphorylation

(127) In *Lathyrus aphaca*, the leaves are modified into

- (A) Spine
- (B) Tendril
- (C) Scale
- (D) Stem-like structure

(128) The movement of auxin is largely

- (A) Basipetal
- (B) Acropetal
- (C) Centripetal
- (D) Centrifugal

(129) Multipolar nerve cells are present in

- (A) Cochlea
- (B) Dorsal root ganglia of spinal cord
- (C) Retina of eye
- (D) Brain

(130) Which of the following is considered the structural and functional unit of neural tissue?

- (A) Myelin sheath
- (B) Axon
- (C) Dendrite
- (D) Neuron

(131) Which of the following statements is wrong for viroids?

- (A) They cause infections.
- (B) Their *RNA* is of high molecular weight.
- (C) They lack a protein coat.
- (D) They are smaller than viruses.

(132) Who proposed the Law of Limiting Factors?

- (A) Blackman
- (B) Hill
- (C) Arnon
- (D) Liebig

(133) Mast cells of connective tissue contain

- (A) Vasopressin and relaxin
- (B) Heparin and histamine
- (C) Heparin and calcitonin
- (D) Serotonin and melanin

(134) The innermost layer of the cortex which shows Casparian thickenings in its cells is called as

- (A) Epidermis
- (B) Endodermis
- (C) Pericycle
- (D) Exodermis

(135) In the gel electrophoresis method

- (A) Fragments of *DNA* can be seen without being stained.
- (B) A natural polymer extracted from seaweed is used as the medium (gel).
- (C) Glowing orange bands of *DNA* can be seen upon exposure to visible light on an ethidium bromide-stained gel.
- (D) The larger the size of the *DNA* fragment in the agarose medium, the farther it will move.

(136) Natural selection where more individuals acquire specific character value other than the mean character value, leads to:

- (A) Directional change
- (B) Disruptive change
- (C) Random change
- (D) Stabilising change

(137) Which of the following sexually transmitted diseases do not specifically affect reproductive organs?

- (A) Genital warts and Hepatitis-B
- (B) Syphilis and Genital herpes
- (C) *AIDS* and Hepatitis-B
- (D) Chlamydiasis and *AIDS*

(138) What is antisense technology?

- (A) When a piece of *RNA* that is complementary in sequence is used to stop expression of a specific gene
- (B) *RNA* polymerase producing *DNA*
- (C) A cell displaying a foreign antigen used for synthesis of antigens
- (D) Production of somaclonal variants in tissue cultures

(139) Select the hormone-releasing Intra-Uterine Devices (IUDs).

- (A) Vaults, *LNG - 20*
- (B) Multiload 375, Progestasert
- (C) Progestasert, *LNG - 20*
- (D) Lippes Loop, Multiload 375

(140) A population in a habitat with limited resources shows four phases of growth. Identify the correct sequence of growth.

- (A) Acceleration → deceleration → Lag phase → asymptote
- (B) Lag phase → acceleration → deceleration → asymptote
- (C) Lag phase → asymptote → deceleration → Acceleration → lag phase
- (D) Acceleration → lag phase → deceleration → asymptote

(141) Given below are two statements: one is labelled as Assertion *A* and the other is labelled as Reason *R*.

Assertion *A*: Endometrium is necessary for implantation of blastocyst.

Reason *R*: In the absence of fertilization, the corpus luteum degenerates that causes disintegration of endometrium.

In the light of the above statements, choose the correct answer from the options given below:

- (A) *A* is false but *R* is true.
- (B) Both *A* and *R* are true and *R* is the correct explanation of *A*.
- (C) Both *A* and *R* are true but *R* is NOT the correct explanation of *A*.
- (D) *A* is true but *R* is false.

(142) Mother and father both have blood group 'A'.

They have two children, one with blood group 'O' and the second one with blood group 'A'. They have:

- (A) mother has homozygotic gene, father has heterozygote $I^A I^A$.
- (B) both are homozygotic ($I^A I^A$).
- (C) mother is heterozygotic ($I^A i$) and father is homozygotic ($I^A I^A$).
- (D) both are heterozygotic ($I^A i$).

(143) Haemophilia is caused due to the lack of:

- (A) *ADH* (B) *AHF*
- (C) *STH* (D) *ACTH*

(144) Which of the following is not a desirable feature of a cloning vector?

- (A) Presence of a marker gene
- (B) Presence of single restriction enzyme site
- (C) Presence of two or more recognition sites
- (D) Presence of origin of replication

(145) When two species with different genetic backgrounds show similarities due to adaptation, this phenomenon is called:

- (A) Divergent evolution
- (B) Microevolution
- (C) Co-evolution
- (D) Convergent evolution

(146) When glucose is the normal energy source in a bacterial cell, which gene transcribes a repressor mRNA?

- (A) *z* (B) *y*
- (C) *a* (D) *i*

(147) The alleles are

- (A) A pair of genes governing a specific character such as tallness or dwarfness
- (B) Multiple forms of genes
- (C) Genes governing eye characters
- (D) Genes present in allosomes

(148) Which was the first successful cloned animal that gained worldwide fame?

- (A) Molly (Sheep) (B) Polly (Sheep)
- (C) Chance (Bull) (D) Dolly (Sheep)

(149) The function of copper ions in copper-releasing *IUDs* is:

- (A) they inhibit gametogenesis
- (B) they make the uterus unsuitable for implantation
- (C) they inhibit ovulation
- (D) they suppress sperm motility and the fertilising capacity of sperms

(150) What is the approximate concentration of CO_2 in the atmosphere (in %)?

- (A) 0.0314 (B) 0.34
- (C) 3.34 (D) 4.0

(151) Which animal has become extinct in India?

- (A) Snow leopard (B) Hippopotamus
- (C) Wolf (D) Cheetah

(152) Gel electrophoresis is used for:

- (A) Construction of recombinant *DNA* by joining with cloning vectors
- (B) Isolation of *DNA* molecules
- (C) Cutting of *DNA* into smaller fragments
- (D) Separation of *DNA* fragments according to their size

(153) In *lac* operon, *z* gene codes for:

- (A) β -galactosidase (B) Permease
- (C) Repressor (D) Transacetylase

(154) In the life cycle of *Plasmodium vivax*, if *a* (Trophozoite), *b* (gametocyte), *c* (oocyst), *d* (zygote), *e* (sporozoite) are different stages. Select the correct sequence starting from the stage it enters the human body.

- (A) $e \rightarrow a \rightarrow b \rightarrow c \rightarrow d$ (B) $e \rightarrow a \rightarrow b \rightarrow d \rightarrow c$
- (C) $c \rightarrow a \rightarrow d \rightarrow b \rightarrow e$ (D) $c \rightarrow b \rightarrow e \rightarrow a \rightarrow d$

(155) Match the following.

I	II	III
(P) Tri-choderma polysporum	(a) statins	(i) clot buster
(Q) Monascus purpureus	(b) streptokinase	(ii) organ transplant
(R) Streptococcus	(c) cyclosporin-A	(iii) cholesterol lowering agent

- (A) $P - (c - iii), Q - (b - i), R - (a - ii)$
- (B) $P - (a - iii), Q - (c - i), R - (b - ii)$
- (C) $P - (b - ii), Q - (a - i), R - (c - iii)$
- (D) $P - (c - ii), Q - (a - iii), R - (b - i)$

(156) The process of removal of anther from the flower bud before it dehisces is called as

- (A) Emasculation
- (B) Bagging
- (C) Embryo rescue
- (D) Budding

(157) Select the *CORRECT* statement.

- (A) Darwin's finches living on the Galapagos islands are different in only color from those found on main land.
- (B) Darwin's finches living on Galapagos island are different in only vegetarian food from those found on main land.
- (C) Darwin's finches living on Galapagos island are different in only shapes of beak.
- (D) Darwin's finches living on Galapagos island are different in size, colour, beak and food habits from those on main land.

(158) In human beings, multiple genes are involved in the inheritance of

- (A) Colour blindness (B) Phenylketonuria
- (C) Sickle-cell anaemia (D) Skin colour

(159) Among the following sets of examples for divergent evolution, select the incorrect option:

- (A) Eye of octopus, bat and man
- (B) Forelimbs of man, bat and cheetah
- (C) Brain of bat, man and cheetah
- (D) Heart of bat, man and cheetah

(160) Human population in India reached the 1 billion mark by the year

- (A) 1995 (B) 1999
- (C) 2000 (D) 2011

(161) What is the study of the interrelationships between living organisms and their environment called?

- (A) Ecology
- (B) Ecosystem
- (C) Phytogeography
- (D) Bioecology

(162) Select the incorrect statement with respect to acquired immunity.

- (A) Anamnestic response is elicited on subsequent encounters with the same pathogen.
- (B) Anamnestic response is due to memory of first encounter.
- (C) Acquired immunity is non-specific type of defense present at the time of birth.
- (D) Primary response is produced when our body encounters a pathogen for the first time.

(163) Which of the following prevents self-pollination?

- (A) Self-sterility
- (B) Herkogamy
- (C) Dichogamy
- (D) All of the above

(164) The theory of natural selection is based on which of the following?

- (A) Contribution of the environment to evolution
- (B) Natural selection acts on favorable variations
- (C) Genetic variations result from changes in the gene pool
- (D) None of the above

(165) Hybridomas are employed for

- (A) Synthesis of antibiotics
- (B) Killing cancer cells
- (C) Synthesis of monoclonal antibodies
- (D) Production of somatic hybrids

(166) Which one of the following cells secretes a hormone?

- (A) Cells of Leydig
- (B) Cells of Sertoli
- (C) Primary spermatocyte
- (D) Secondary spermatocyte

(167) Match the ecological terms given in Column I with its correct explanation in Column II.

Column I (Terms)	Column II (Explanation)
1. Stratification	a. Horizontal distribution of different species
2. Zonation	b. Rate of generation of Biomass in an ecosystem
3. Productivity	c. Breakdown of detritus
4. Decomposition	d. Vertical distribution of different species

Choose

the correct option.

- (A) $1-d, 2-c, 3-a, 4-b$ (B) $1-d, 2-a, 3-b, 4-c$
- (C) $1-c, 2-d, 3-a, 4-b$ (D) $1-c, 2-a, 3-d, 4-b$

(168) In an ecosystem, 1000 J of energy is stored at the producer level. How much energy will be available at the secondary consumer level (in J)?

- (A) 10 (B) 0.1
- (C) 100 (D) 1

(169) One of the following is not a method of contraception. Which one?

- (A) Condoms
- (B) Pills of a combination of oxytocin and vasopressin
- (C) Lippes loop
- (D) Tubectomy

(170) The yield of a paddy field can be increased by the application of:

- (A) Iron bacteria (B) Nostoc/Anabaena
- (C) Archaeobacteria (D) Symbiotic bacteria

(171) A complex of ribosomes attached to a single strand of RNA is known as

- (A) polypeptide (B) Okazaki fragment
- (C) polysome (D) polymer

(172) In case of snakebites, the injection given to patients contains preformed antibodies against the snake venom. This type of immunisation is called

- (A) Active immunity
- (B) Both kinds of immunity
- (C) Passive immunity
- (D) Partial passive immunity

(173) Which of the following layers in an antral follicle is acellular?

- (A) Stroma (B) Zona pellucida
- (C) Granulosa (D) Theca interna

(174) In large cities, domestic sewage

- (A) has high BOD as it contains both aerobic and anaerobic bacteria.
- (B) is processed in sewage treatment plants through primary aerobic and then secondary anaerobic processes.
- (C) does not require aeration when processed in STP because it contains sufficient oxygen.
- (D) contains large amounts of suspended solids and dissolved salts.

(175) In relation to Gross primary productivity and Net primary productivity of an ecosystem, which one

of the following statements is correct?

- (A) There is no relationship between Gross primary productivity and Net primary productivity.
- (B) Gross primary productivity is always less than net primary productivity.
- (C) Gross primary productivity is always more than net primary productivity.
- (D) Gross primary productivity and Net primary productivity are one and same.

(176) Unisexuality of flowers prevents

- (A) geitonogamy, but not xenogamy
- (B) autogamy and geitonogamy
- (C) autogamy, but not geitonogamy
- (D) both geitonogamy and xenogamy.

(177) Which of the following is incorrect regarding vasectomy?

- (A) No sperm occurs in seminal fluid
- (B) No sperm occurs in epididymis
- (C) Vasa deferentia is cut and tied
- (D) Irreversible sterility

(178) What is true about cleavage in the fertilised egg in humans?

- (A) It is meroblastic.
- (B) It starts while the egg is in the fallopian tube.
- (C) It is identical to normal mitosis.
- (D) It starts when the egg reaches the uterus.

(179) Triploid plants can be obtained from the culture of:

- (A) Pollen (B) Endosperm
- (C) Ovule (D) Megaspore

(180) Which of the following is incorrectly matched regarding the microbe, its derivative, and its application?

- (A) *Monascus purpureus* - Statins - Lowers blood cholesterol levels.
- (B) *Streptococcus* - Streptokinase - Removes clots from blood vessels.
- (C) *Clostridium butylicum* - Lipase - Used to remove oily stains in laundry.
- (D) *Trichoderma polysporum* - Cyclosporin-A - Immunosuppressive agent in organ transplant patients.